

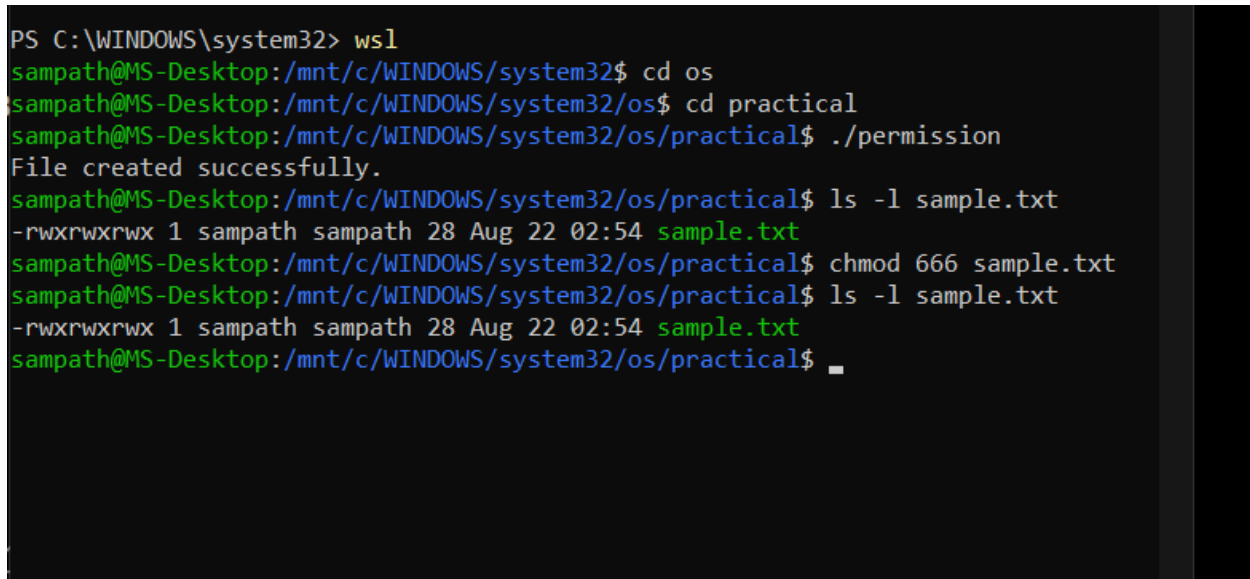
File Permission



The screenshot shows a code editor window titled "permission.c" with the file path "/mnt/c/WINDOWS/system32/os/practical". The code is as follows:

```
1 #include <stdio.h>
2 int main()
3 {
4     FILE *fp;
5     fp = fopen("sample.txt", "w");
6     if (fp == NULL)
7     {
8         printf("Cannot open file\n");
9         return 1;
10    }
11    fprintf(fp, "Hello, this is a test file.\n");
12    fclose(fp);
13    printf("File created successfully.\n");
14    return 0;
15 }
16 |
```

- 1) Read and Write only to the owner



The screenshot shows a terminal window with the following commands and output:

```
PS C:\WINDOWS\system32> wsl
sampath@MS-Desktop:/mnt/c/WINDOWS/system32$ cd os
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os$ cd practical
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ ./permission
File created successfully.
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ ls -l sample.txt
-rwxrwxrwx 1 sampath sampath 28 Aug 22 02:54 sample.txt
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ chmod 666 sample.txt
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ ls -l sample.txt
-rwxrwxrwx 1 sampath sampath 28 Aug 22 02:54 sample.txt
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$
```

- 2) Read and Write to the owner and group

```

-rwxrwxrwx 1 sampath sampath 28 Aug 22 02:54 sample.txt
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ chmod 660 sample.txt
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ ls -l sample.txt
-rwxrwxrwx 1 sampath sampath 28 Aug 22 02:54 sample.txt
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$

```

3) Only owner can read and write

```

sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ chmod 600 sample.txt
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ ls -l sample.txt
-rwxrwxrwx 1 sampath sampath 28 Aug 22 02:54 sample.txt

```

4) Owner read/write, everyone else read-only

```

sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ chmod 644 sample.txt
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ ls -l sample.txt
-rwxrwxrwx 1 sampath sampath 28 Aug 22 02:54 sample.txt
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$

```

5) Owner can do everything, others can only read and execute

```

sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ chmod 755 sample.txt
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ ls -l sample.txt
-rwxrwxrwx 1 sampath sampath 28 Aug 22 02:54 sample.txt

```

6) Owner and group can read/write

```

sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ chmod 660 sample.txt
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ ls -l sample.txt
-rwxrwxrwx 1 sampath sampath 28 Aug 22 02:54 sample.txt

```

7) Only owner can execute

```

sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ chmod 100 sample.txt
sampath@MS-Desktop:/mnt/c/WINDOWS/system32/os/practical$ ls -l sample.txt
-r-xr-xr-x 1 sampath sampath 28 Aug 22 02:54 sample.txt

```

8) Owner: read + write; Group: nothing; Others: nothing → 600

9) Owner: read + write; Group: read; Others: read → 644

10) Owner: read + write + execute; Group: read + execute; Others: read + execute → 755

11) Owner: read + write; Group: read + write; Others: nothing → 660

12) Everyone: read only → 444